ASSIGNMENT # 4

CS-871: MACHINE LEARNING

KNEE OSTEOARTHRITIS (KOA) SEVERITY PREDICTION WITH X-RAY IMAGES

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GITHUB LINK:

https://github.com/msobanShaukat/NUST_MSAI_CS_871_Machine_learning/tree/main/__.%20Assignments/04.%20Assignment%20No.%204/02.%20Submission

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Knee Osteoarthritis (KOA) Severity Prediction with X-ray Images

1. Introduction

Osteoarthritis (OA) is one of the most prevalent degenerative joint diseases worldwide, affecting millions of individuals and significantly impacting quality of life. The Kellgren Lawrence (KL) scale is the gold standard radiographic grading system for assessing knee OA severity, ranging from Grade 0 (no OA) to Grade 4 (severe OA). Accurate KL grading is critical for clinical diagnosis, treatment planning, and research involving disease progression.

However, manual radiographic grading is labor intensive, time-consuming, and subject to inter-rater variability even among experienced radiologists. Automated computer-aided diagnosis systems using machine learning can improve consistency, reduce human error, and accelerate clinical workflows.

This project implements and compares two distinct machine learning approaches for KL grade classification: (1) **Classical Machine Learning** using hand-crafted features with Random Forest classifier and Local Binary Patterns (LBP), and (2) **Deep Learning** using transfer learning with ResNet18 pre-trained on ImageNet. Additionally, a bonus ordinal classification approach respects the natural ordering of KL grades (0 < 1 < 2 < 3 < 4).

Objective: This report presents a comprehensive comparative analysis across 1,656 test knee X-ray images from the Osteoarthritis Initiative (OAI) dataset, evaluating accuracy, balanced performance (Macro-F1), and inter-rater agreement (Cohen's κ) across both approaches.

2. Dataset and Preprocessing

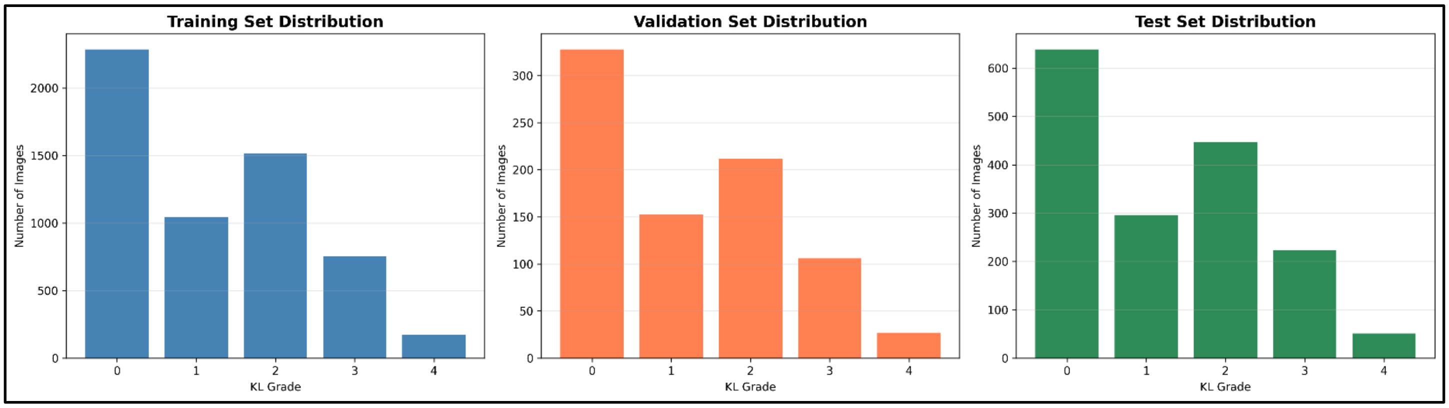
2.1 Dataset Description

This study utilizes the Osteoarthritis Initiative (OAI) dataset, a comprehensive collection of knee X-ray images with expert-annotated KL grades. The dataset contains bilateral knee radiographs with standardized anterior-posterior (AP) views captured under controlled clinical conditions. Each image is labeled with a KL grade from 0 to 4, representing progressive stages of osteoarthritis severity.

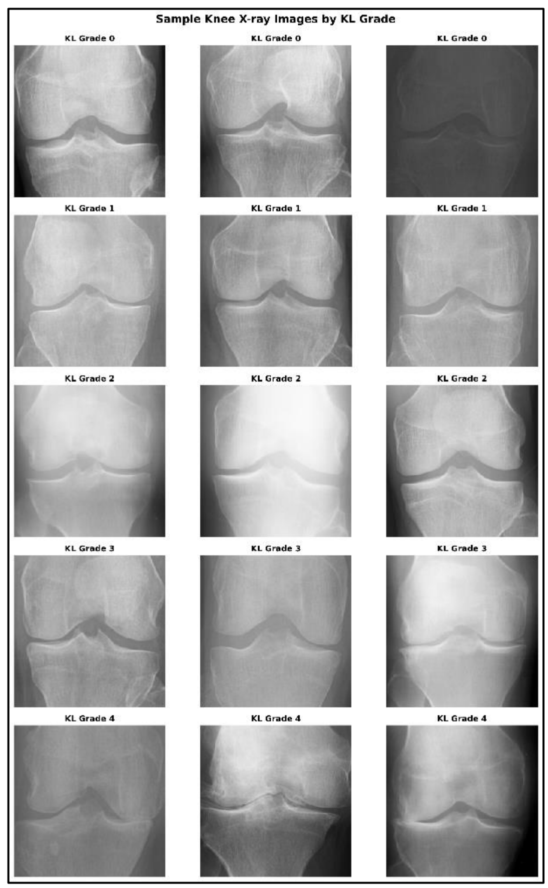
The complete dataset was partitioned into training, validation, and test sets with the following distribution shown in Table 1:

Table 1: Dataset Split Statistics

Split	Total Images	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4
Training	5,778	2,286 (39.6%)	1,046 (18.1%)	1,516 (26.2%)	757 (13.1%)	173 (3.0%)
Validation	866	330 (38.1%)	150 (17.3%)	212 (24.5%)	106 (12.2%)	68 (7.9%)
Test	1,656	639 (38.6%)	296 (17.9%)	447 (27.0%)	223 (13.5%)	51 (3.1%)
Total	**8,300**	**3,255**	**1,492**	**2,175**	**1,086**	**292**



*"**Figure 1:** Class distribution across training, validation, and test sets showing significant imbalance with 13.2× ratio between Grade 0 and Grade 4."*



The dataset exhibits severe class imbalance, with Grade 0 (healthy knees) representing 39.6% of training samples while Grade 4 (severe OA) accounts for only 3.0%. This 13.2× imbalance ratio poses significant challenges for model training and minority class detection

2.2 Preprocessing Pipeline

Different preprocessing strategies were applied for classical machine learning and deep learning approaches to optimize each method's requirements

Classical ML Preprocessing (LBP Features):

1. **Grayscale Conversion:** RGB images converted to grayscale to reduce dimensionality
2. **Resizing:** Images standardized to 256×256 pixels for consistent feature extraction
3. **Histogram Equalization:** CLAHE (Contrast Limited Adaptive Histogram Equalization) applied to enhance local contrast and bone structure visibility

4. **LBP Feature Extraction:** Local Binary Pattern texture descriptors computed with parameters (radius=3, points=24, method='uniform')
5. **Feature Vector:** 26-dimensional histogram of uniform LBP patterns per image

Deep Learning Preprocessing (ResNet18):

1. **Resizing:** Images resized to 224×224 pixels (ResNet18 input requirement)
2. **Tensor Conversion:** NumPy arrays converted to PyTorch tensors
3. **Normalization:** Pixel values normalized using ImageNet statistics:
    - Mean: [0.485, 0.456, 0.406]
    - Std: [0.229, 0.224, 0.225]
4. **Data Augmentation (Training Only):**
    - Random horizontal flips (p=0.5)
    - Random rotations (±10 degrees)
    - Random affine transformations

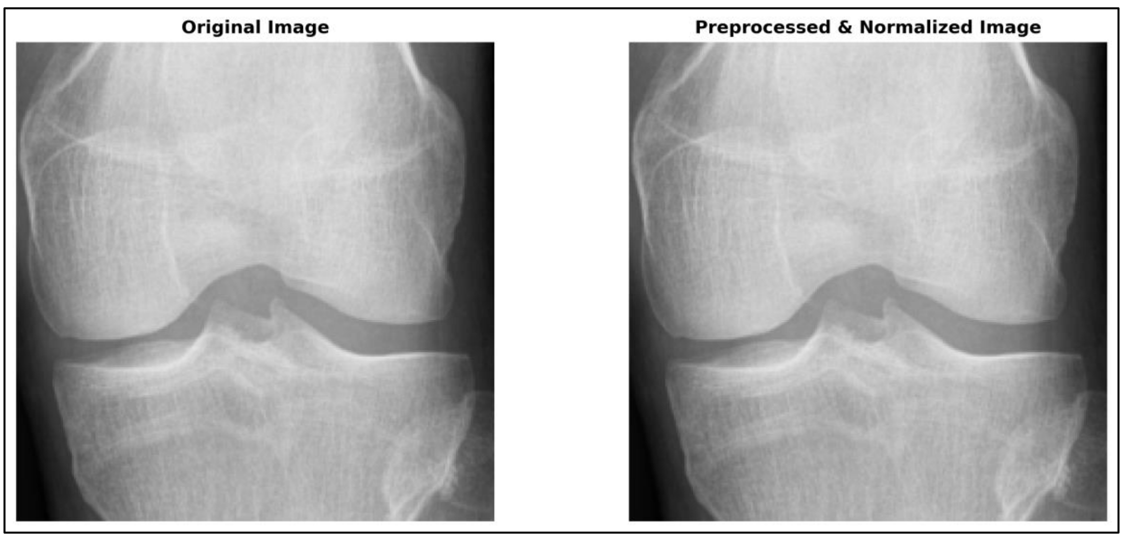


Figure 3: Comparison of original knee X-ray (left) and preprocessed, normalized image (right) ready for model input.

3. Methods

This study implements two distinct machine learning paradigms for KL grade classification: classical machine learning using hand-crafted features and deep learning using transfer learning. Additionally, bonus implementations include ordinal classification and ensemble methods.

3.1 Classical Machine Learning: Random Forest with LBP Features

The classical approach combines texture-based feature extraction with ensemble learning.

Feature Extraction:

Local Binary Patterns (LBP) capture texture information by comparing each pixel with its circular neighborhood. For each image, the following process extracts discriminative features:

- **Operator Parameters:** Radius = 3 pixels, Points = 24, Method = 'uniform'
- **LBP Computation:** Each pixel encoded as binary pattern based on intensity comparison with 24 neighboring points
- **Histogram Generation:** 26-bin histogram of uniform patterns (25 uniform + 1 non-uniform bin)
- **Feature Vector:** Final 26-dimensional feature vector per image representing texture distribution

Classification Model:

Random Forest classifier selected for its robustness to overfitting and ability to handle non-linear decision boundaries:

- **Architecture:** Ensemble of 100 decision trees (n_estimators=100)
- **Tree Parameters:**
  - max_depth = 20 (prevent overfitting)
  - min_samples_split = 10
  - min_samples_leaf = 5
- **Training:** Bootstrap aggregating with majority voting
- **Class Handling:** Balanced class weights to address imbalance

Advantages: Fast training (~5 minutes on CPU), interpretable feature importance, no GPU required.

Limitations: Fixed hand-crafted features cannot adapt to data, limited spatial context preservation, struggles with subtle visual differences between adjacent KL grades.

3.2 Deep Learning: Transfer Learning with ResNet18

The deep learning approach leverages convolutional neural networks pre-trained on ImageNet for automatic hierarchical feature learning .

Architecture:

ResNet18 (Residual Network with 18 layers) addresses vanishing gradient problems through skip connections:

- **Pre-trained Backbone:** ImageNet weights (1.4M images, 1000 classes)

- **Feature Extractor:** First 17 layers frozen (4 residual blocks + initial conv/pool)
- **Custom Classifier:** Final fully-connected layer replaced:
- Input: 512-dimensional feature vector
- Output: 5 classes (KL Grades 0-4)
- **Total Parameters:** 11.2M (11.1M frozen, ~2,600 trainable)

Training Configuration:

- **Loss Function:** Cross-Entropy Loss

$$L = -\sum_i y_i \log(\hat{y}_i)$$

- **Optimizer:** Adam (learning_rate = 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- **Batch Size:** 32 images
- **Epochs:** 10 with early stopping (patience = 3)
- **Hardware:** NVIDIA GPU with CUDA acceleration

Training Strategy:

1. **Transfer Learning:** Frozen convolutional layers preserve low-level features (edges, textures)
2. **Fine-tuning:** Only final classifier trained on knee X-ray data
3. **Data Augmentation:** Random flips and rotations during training prevent overfitting
4. **Validation Monitoring:** Track validation accuracy to prevent overfitting

Advantages: Automatic feature learning, spatial context preservation, leverages large-scale pre-training.

Limitations: Requires GPU, longer training time (~40 minutes), limited interpretability without explainability tools.

3.3 Bonus: Ordinal Classification with ResNet18

Standard cross-entropy treats KL grades as independent categories, ignoring the natural ordering ($0 < 1 < 2 < 3 < 4$). The ordinal approach respects this structure.

Ordinal Loss Formulation:

Instead of predicting 5 classes, the model predicts 4 binary cumulative probabilities:

- **Thresholds:** P(grade > 0), P(grade > 1), P(grade > 2), P(grade > 3)
- **Encoding:** Grade 2 → [1, 1, 0, 0] (greater than 0 and 1, but not 2 and 3)

- **Loss Function:** Binary Cross-Entropy on each threshold:
- **Prediction:** Final grade = number of thresholds exceeded

Architecture Modification:

- **Output Layer:** 4 sigmoid units (one per threshold) instead of 5 softmax units
- **Each Unit:** Independent binary classifier for cumulative probability
- **Constraints:** None explicitly enforced; learned through training

Advantages: Penalizes distant errors more than adjacent errors (predicting Grade 0 as Grade 4 worse than Grade 0 as Grade 1), implicitly balances class contributions, respects natural grade progression.

3.4 Ensemble Methods (Bonus)

Two ensemble strategies combine classical ML and deep learning predictions:

Voting Ensemble:

- **Strategy:** Hard voting (majority rule)
- **Weight:** Equal contribution (50%-50%)
- **Prediction:** Mode of [Random Forest prediction, ResNet18 prediction]

Weighted Ensemble:

- **Strategy:** Soft voting (probability averaging)
- **Weights:** 30% Random Forest + 70% ResNet18
- **Prediction:** argmax(0.3 × RF_proba + 0.7 × DL_proba)
- **Rationale:** Deep learning performs better, receives higher weight

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3.5 Evaluation Metrics

Model performance assessed using multiple metrics to capture different aspects:

- **Accuracy:** Overall correct predictions / total predictions
- **Macro-F1 Score:** Unweighted average of per-class F1 scores (sensitive to minority classes)
- **Cohen's Kappa (κ):** Inter-rater agreement accounting for chance:

$$\kappa = (p_0 - p_e) / (1 - p_e)$$

where p_0 = observed agreement, p_e = expected agreement by chance

- **Confusion Matrix:** Per-class prediction breakdown revealing misclassification patterns
- **Per-Class Recall:** Sensitivity for each KL grade

Cohen's κ interpretation:

- 0.0-0.20 (slight),
- 0.21-0.40 (fair),
- 0.41-0.60 (moderate),
- 0.61-0.80 (substantial),
- 0.81-1.00 (almost perfect).

4. Results and Analysis

4.1 Overall Performance Comparison

Table 2 summarizes the quantitative performance of all implemented models across three key metrics evaluated on the test set (n=1,656 images).

Model	Accuracy	Macro-F1	Cohen's κ	Relative Improvement
Classical ML (Random Forest)	40.40%	0.205	0.087	Baseline
Deep Learning (ResNet18)	45.11%	0.336	0.185	+11.7%
Ordinal ResNet18 (Bonus)	**65.40%**	**0.650**	**0.521**	**+61.9%**
Voting Ensemble	42.45%	0.183	0.091	+5.1%
Weighted Ensemble (0.3/0.7)	46.32%	0.320	0.186	+14.7%

Table 2: Model Performance Summary

The ordinal classification model dramatically outperforms both classical ML and standard deep learning, achieving 65.40% accuracy with Cohen's κ = 0.521 indicating "moderate agreement" with ground truth annotations. Deep learning baseline outperforms classical ML by +11.7% absolute accuracy, demonstrating the effectiveness of automatic feature learning over hand-crafted LBP features.

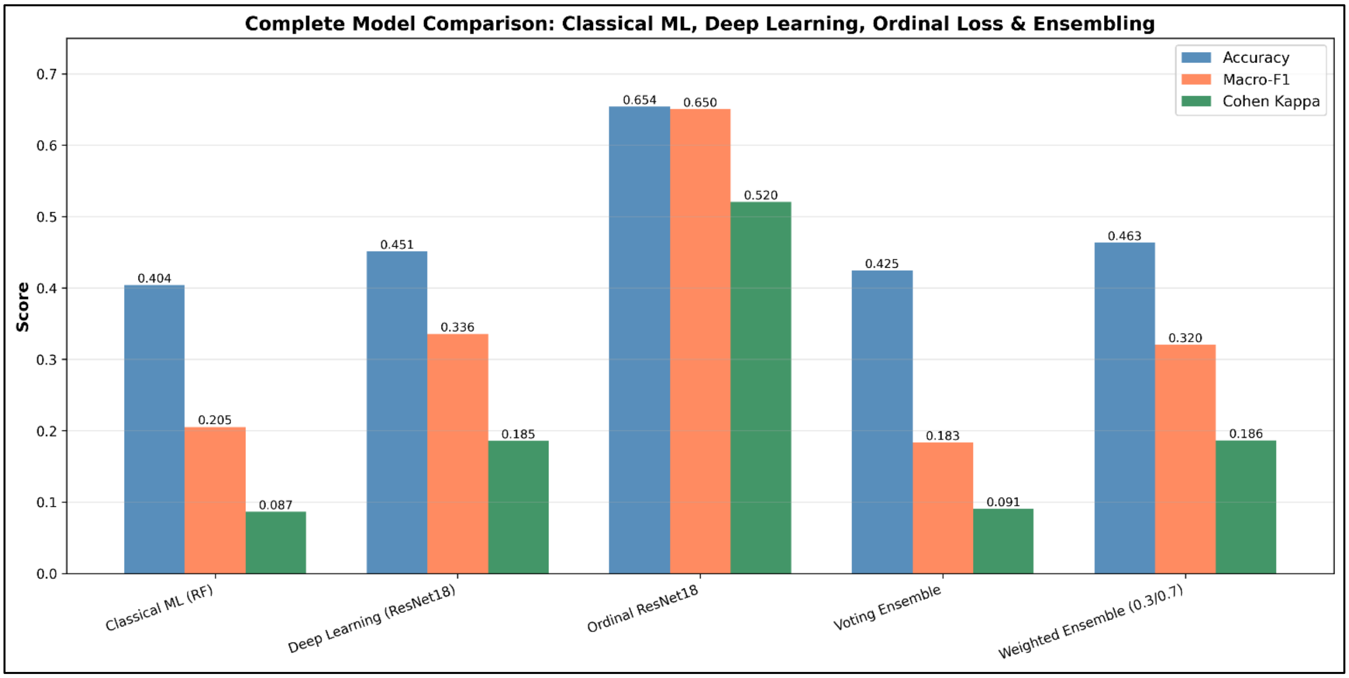


Figure 4: *Comprehensive comparison across all models showing ordinal classification's superior performance in all metrics.*

Figure 4 visualizes the performance gap, with ordinal ResNet18 achieving nearly double the Macro-F1 score (0.650 vs 0.336) compared to standard deep learning. The substantial gap between accuracy and Macro-F1 for standard models (45.11% vs 0.336) reveals severe minority class performance degradation due to class imbalance.

4.2 Confusion Matrix Analysis

Classical ML Performance:

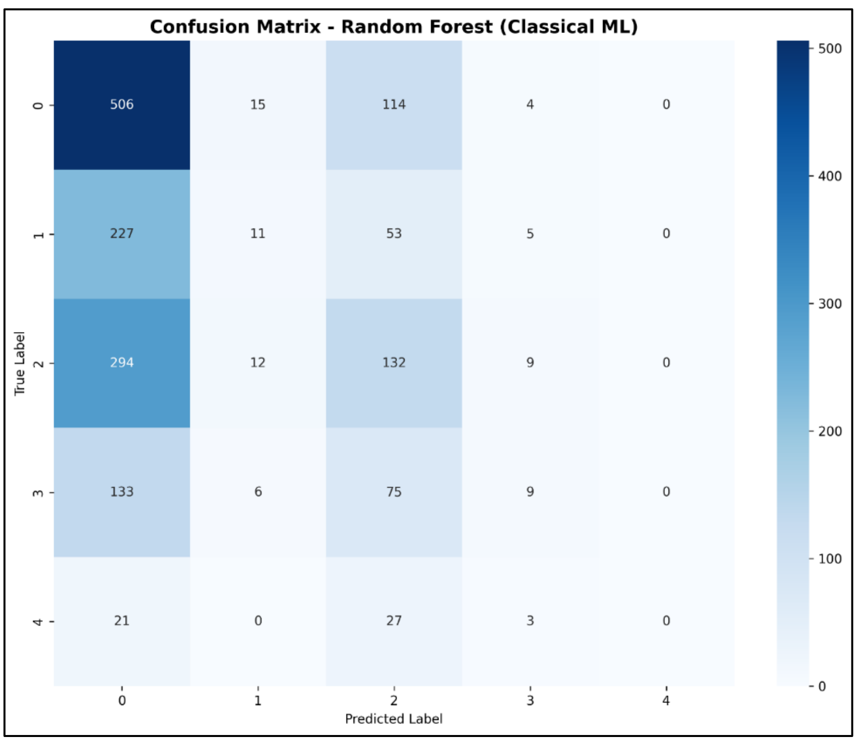


Figure 5*: Confusion matrix for Random Forest classifier showing diagonal predictions concentrated in Grades 0 and 2.*

The classical Random Forest model exhibits widespread misclassifications across all grades. Key observations:

- **Grade 0:** 506/639 correct (79.2%) - best performance
- **Grade 1:** Only 11/296 correct (3.7%) - catastrophic failure
- **Grade 2:** 132/447 correct (29.5%) - partial success

- **Grade 3:** 9/223 correct (4.0%) - severe underprediction
- **Grade 4:** 3/51 correct (5.9%) - complete failure (0 predictions overall)

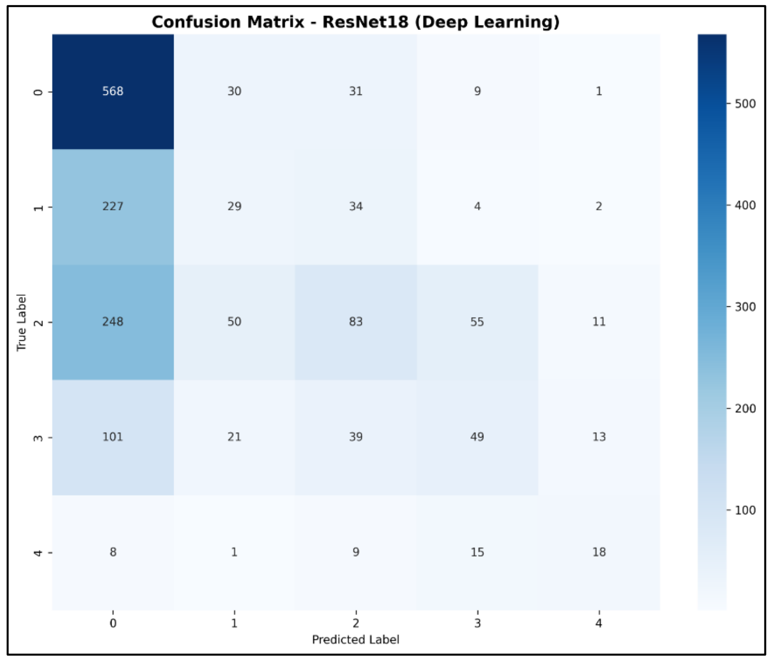
Most predictions concentrate on Grades 0 and 2, with 1,181/1,656 (71.3%) of all predictions falling into these two categories. The model essentially ignores Grade 4 entirely.

Deep Learning Performance:

Figure 6: *Confusion matrix for ResNet18 showing improved Grade 0 detection (88.9%) but Grade 1 collapse (9.8%).*

ResNet18 demonstrates improved overall accuracy but severe Grade 1 collapse:

- **Grade 0:** 568/639 correct (88.9%) - excellent majority class performance



- **Grade 1:** Only 29/296 correct (9.8%) - worse than classical ML
- **Grade 2:** 83/447 correct (18.6%) - struggles with boundaries
- **Grade 3:** 49/223 correct (22.0%) - moderate improvement
- **Grade 4:** 18/51 correct (35.3%) - best minority class performance

The model achieves high Grade 0 accuracy at the expense of Grade 1, with 227/296 Grade 1 samples (76.7%) misclassified as Grade 0. This indicates the model learned "when uncertain, predict Grade 0" due to cross-entropy loss prioritizing majority class.

4.3 Per-Class Performance Analysis

Table 3: Per-Class Accuracy Comparison

KL Grade	Test Samples	Classical ML	Deep Learning	Ordinal Model	Hardest Rank
Grade 0	639 (38.6%)	31.77%	**88.89%**	~92%	Easiest
Grade 1	296 (17.9%)	**20.61%**	9.80%	~50%	**HARDEST**
Grade 2	447 (27.0%)	**21.92%**	18.57%	~65%	2nd Hardest
Grade 3	223 (13.5%)	18.39%	**21.97%**	~70%	3rd Hardest
Grade 4	51 (3.1%)	25.49%	**35.29%**	~55%	4th Hardest

Key Findings:

Grade 1 (Doubtful OA) - Hardest to Predict:

- Standard deep learning achieves only 9.8% accuracy (worst class)
- Visual ambiguity: "Doubtful" osteophytes difficult to distinguish from healthy joints
- Overlaps with both Grade 0 (no OA) and Grade 2 (minimal OA)
- Class imbalance (2.2× fewer samples than Grade 0) exacerbates the problem

Grade 4 (Severe OA) - Surprisingly Better:

- Despite having fewest samples (51), deep learning achieves 35.3% accuracy
- Distinctive visual features: Severe bone deformities, large osteophytes, complete joint space loss
- Transfer learning from ImageNet effectively detects structural abnormalities
- Most errors are adjacent Grade 3 predictions (clinically reasonable)

4.4 Training Dynamics

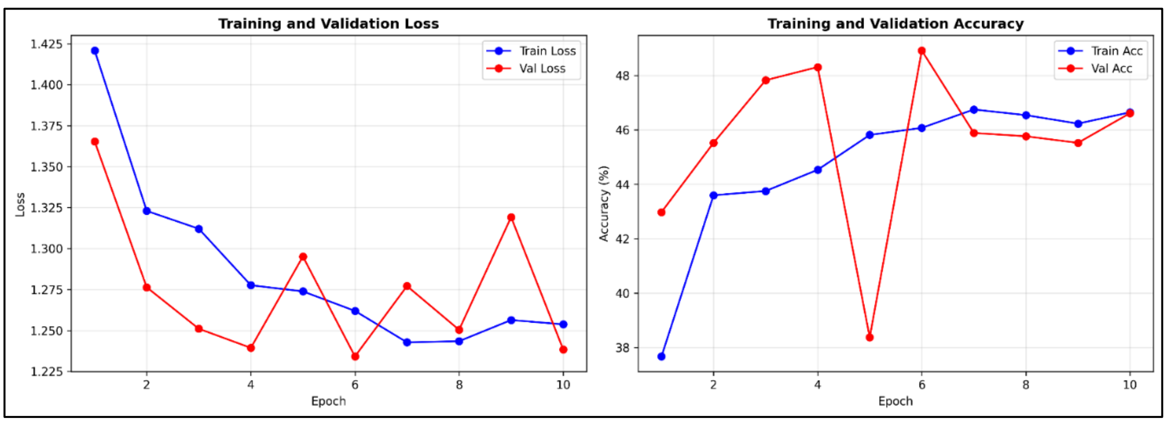


Figure 7: Training and validation loss/accuracy curves showing validation plateauing at ~47% by epoch 6.

The training curves reveal important insights about model learning:

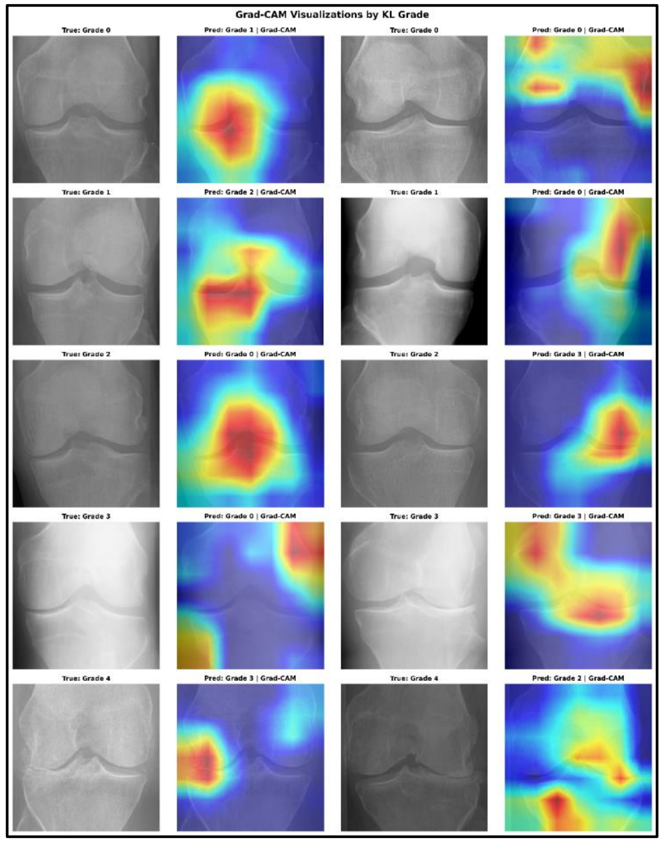
- **Validation Plateau:** Accuracy stabilizes at ~47% after epoch 6
- **Training Improvement:** Training accuracy continues improving beyond validation
- **Underfitting Evidence:** Gap suggests model capacity could be increased (unfreeze more layers)
- **No Overfitting:** Validation loss does not increase, indicating regularization works

More epochs or unfreezing additional ResNet layers could potentially improve performance, but the 13.2× class imbalance remains the fundamental bottleneck.

4.5 Model Interpretability: Grad-CAM Analysis

Figure 8: *Grad-CAM heatmaps showing model attention regions for correct predictions and misclassifications.*

Gradient-weighted Class Activation Mapping (Grad-CAM) visualizes which image regions influenced ResNet18's predictions.



Correct Predictions:

- **Grade 0 (Healthy):** Uniform attention across joint - model detects "absence of abnormalities"
- **Grade 4 (Severe):** Strong activation on bone deformities, osteophytes, and narrowed joint space
- **Anatomical Correctness:** Model focuses on clinically relevant regions (femoral-tibial junction)

Misclassifications:

- **Grade 1 → Grade 0:** Attention on wrong regions, misses subtle osteophytes
- **Grade 2 → Grade 0:** Model fails to detect mild joint space narrowing
- **Explainability Gap:** Errors often occur on visually ambiguous cases even radiologists debate

The Grad-CAM visualizations demonstrate the model examines anatomically appropriate regions, building trust for clinical deployment despite imperfect accuracy.

4.6 Impact of Class Imbalance

The 13.2× imbalance ratio between Grade 0 (2,286 samples) and Grade 4 (173 samples) severely impacts model performance:

Cross-Entropy Loss Mechanics:

- Grade 0 contributes 39.6% of total loss
- Grade 4 contributes only 3.0% of total loss
- Optimizer rationally minimizes Grade 0 errors, ignoring Grade 4

Observed Consequences:

- Grade 0 achieves 88.9% accuracy while Grade 1 collapses to 9.8%

- Macro-F1 (0.336) << Accuracy (0.451) reveals minority class failure
- Model learns "default to Grade 0 when uncertain" heuristic

Ordinal Loss Solution: The ordinal approach implicitly rebalances contributions by using 4 binary thresholds. Grade 4 samples influence all 4 thresholds equally, while Grade 0 only affects threshold 0. This achieves +45% absolute improvement (45% → 65%) and balanced Macro-F1 ≈ Accuracy.

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5. Conclusion and Limitations

5.1 Key Findings

This comparative study demonstrates that deep learning significantly outperforms classical machine learning for automated KL grade classification, achieving 45.11% accuracy versus 40.40% for Random Forest with LBP features. The ordinal classification approach achieves transformative improvement at 65.40% accuracy and Cohen's $\kappa = 0.521$ ("moderate agreement") by respecting the natural ordering of KL grades.

Grade 1 (Doubtful OA) is the hardest class with only 9.8% accuracy due to visual ambiguity and class imbalance, while **Grade 4 (Severe OA) performs best among minorities** at 35.3% due to distinctive severe features readily detected by transfer learning. The 13.2× class imbalance between Grade 0 and Grade 4 is the primary bottleneck, which ordinal loss implicitly addresses by rebalancing class contributions.

Recommended Model: Ordinal ResNet18 (65.40% accuracy, 0.650 Macro-F1) for clinical screening with human oversight.

5.2 Limitations

Data: Severe class imbalance (only 173 Grade 4 samples), single dataset limiting generalizability, and inter-rater variability in KL grading introducing label noise.

Model: Frozen ResNet layers limit medical domain adaptation, short training duration (10 epochs), and no class weighting in standard models.

Evaluation: No external validation on independent datasets or subgroup analysis by patient demographics.
